

PROBLEM STATEMENT #39 | Retail, E-Commerce & Finance

Second-Hand Product Escrow Marketplace

1. Problem Context & Overview

A secure peer-to-peer trading marketplace where buyer payments are held in escrow, independent third-party inspectors certify product condition, and funds release only upon buyer sign-off.

Target Stakeholders / Actors: *Buyer, Inspection Agent*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall lock buyer payment in escrow and transition order state to 'Under Inspection' until an inspector submits physical verification sign-off.

Sample Acceptance Criteria: Pass: Funds held securely and released only on dual verification; Fail: Funds released prematurely to seller.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: Dispute resolution tickets must freeze escrow funds immediately and prevent account withdrawals during active investigation.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.